

PROFILE REPORT

Hirok Roy Rahul

Aspiring AI Researcher & Full-Stack Developer

BRAC University, Dhaka, Bangladesh

Compiled: May 2026

1. Personal Profile

Full Name	Hirok Roy Rahul
Known As	Hirok
Online Handles	hirokr / hrrahul / hrrahul0625
Location	Lalmonirhat, Rangpur Division, Bangladesh
University Base	Dhaka, Bangladesh (BRAC University)
Email	work.hrr25@gmail.com
AUV Team Email	hrahul@bracu-duburi.com
Portfolio	hirokr.vercel.app
LinkedIn	linkedin.com/in/hirokr
GitHub	github.com/hirokr
ORCID	0009-0009-3782-5632
ResearchGate	researchgate.net/profile/Hirok-Roy-Rahul
Kaggle	kaggle.com/hirokr
LeetCode	leetcode.com/hrrahul
Codeforces	codeforces.com/profile/hrrahul0625
HackerRank	hackerrank.com/hirokr
First Language	Bengali (English is the second language)

2. Education

BSc in Computer Science & Engineering

Institution: BRAC University, Dhaka, Bangladesh

Expected Graduation: December 2026 (Final Year)

Year: 4th Year (Final Year Undergraduate)

Relevant coursework includes data structures, algorithms, computer networks (Cisco Packet Tracer), operating systems, databases, computer vision, machine learning, software engineering, and non-departmental courses in Economics. Active participation in the BRACU DUBURI AUV research team as part of his academic journey.

3. Professional Experience

Frontend Developer — FourbTech

Employed as a frontend developer at FourbTech, a startup, where Hirok applies his React/Next.js skills in a production environment. This role runs concurrently with his final year of studies, giving him real-world exposure to product development, client-facing UI work, and collaborative engineering workflows.

ML Research Assistant — BioSE Research Center

Serves as a Machine Learning Research Assistant at the BioSE Research Center, contributing to biomedical AI research. This role directly aligns with his academic interests in computer vision and medical AI, providing hands-on research experience that feeds into his long-term goal of publishing in high-impact journals.

Machine Vision Engineer — BRACU DUBURI AUV Team

Member of the Machine Vision subteam for BRACU DUBURI, Bangladesh's first Autonomous Underwater Vehicle. The team competes in international robotics competitions including RoboSub. His responsibilities include developing computer vision algorithms for underwater object detection, gate recognition, and navigation — high-stakes real-time engineering work requiring expertise in OpenCV, Python, and embedded systems.

MLE Contractor Applications

Applied for a Machine Learning Engineer contractor role at Turing and an Operations & AI Automation Specialist role at Camel Tech, indicating active pursuit of professional ML/AI work beyond academia.

4. Technical Skills & Competencies

Programming Languages

- Python (ML/CV, automation, TensorFlow, PyTorch, pandas, NumPy, OpenCV, spaCy)
- TypeScript / JavaScript (Next.js, Node.js, React, LangChain JS, BullMQ)
- C / C++ (multithreading, OS-level programming, embedded systems)
- Java (object-oriented programming, academic use)
- SQL (MySQL, PostgreSQL via Prisma ORM)
- Bash / Shell scripting
- MATLAB (academic and signal processing use)

Frameworks & Libraries

- Frontend: Next.js 14, React, React Native, Expo, Three.js / React Three Fiber, Tailwind CSS
- Backend: Node.js, Express, FastAPI, Django
- AI/ML: TensorFlow, PyTorch, LangChain, EfficientNet, OpenCV, spaCy
- Database & ORM: Prisma, Supabase, MongoDB, MySQL, PostGIS
- Job Queuing: BullMQ
- Real-time & Storage: UploadThing, Mapbox, Capacitor

DevOps & Tools

- Version control: Git / GitHub (68 public repositories; Pair Extraordinaire, Pull Shark x2, YOLO, Quickdraw achievements)
- Cloud: Azure (academic), AWS (pursuing ML Associate certification)
- Design: Figma, Adobe Illustrator, Adobe Photoshop, Blender, Unreal Engine
- Networking: Cisco Packet Tracer (VLSM, RIP v2, DHCP relay, floating static routes)
- Containers & Deployment: Docker, Vercel, zero-budget/free-tier infrastructure expertise

Emerging & Specialized Skills

- Cryptography: Hand-rolled RSA-2048, ECC secp256k1, HMAC, SHA-256 using BigInt (no external libs)
- LLM Engineering: LangChain JS (completed course), streaming LLM outputs, prompt engineering, AI agent design
- Explainable AI (XAI): Cross-attention weights as built-in XAI, attention visualization, counterfactual reasoning
- Computer Vision: Object detection, underwater AUV vision, deepfake detection, skin lesion classification
- AI APIs: Claid AI, Tripo3D, Pixazo (Hunyuan 3D model generation), Groq / Llama 3
- Workflow automation: n8n, Python automation, form-to-spreadsheet-to-cloud pipelines
- Arduino / Embedded systems (AUV hardware-software integration)

5. Active & Notable Projects

Tryora — AI-Powered Virtual Try-On Platform

Stack: TypeScript, Next.js, Node.js, BullMQ, Prisma, Claid AI, Tripo3D/Pixazo, UploadThing

An e-commerce platform that allows users to virtually try on clothing or products using AI-driven image processing and 3D model generation. Demonstrates advanced integration of async job queues, third-party AI APIs, and full-stack product thinking.

- TypeScript/Node.js backend with BullMQ for async job management
- Next.js frontend with clean product UI
- Prisma ORM for database management
- Claid AI for image enhancement; Tripo3D / Pixazo for 3D model generation
- UploadThing for media management

Hangout Buddy — University Social Coordination App

Stack: Next.js 14, Supabase, Mapbox, Capacitor

A mobile-first social app for BRAC University students to coordinate meetups and social events. Notable design decision: no individual GPS coordinates stored, protecting user privacy while still enabling location-aware features.

- Next.js 14 with Capacitor for cross-platform mobile deployment
- Supabase for backend-as-a-service (auth, realtime, database)
- Mapbox integration for location-based features
- Hard constraint: zero individual GPS coordinate storage (privacy by design)

DengueScope — Climate-Driven Disease Outbreak Prediction Platform

Stack: ML (climate + epidemiological data fusion), React/Next.js frontend

Built for the NRF26 National Hackathon. A platform that predicts dengue fever outbreak risk in Bangladesh by correlating climate data (rainfall, temperature, humidity) with historical disease incidence patterns.

- Disease outbreak risk prediction using climate-epidemiological ML models
- Targeted specifically at Bangladesh's geographic and seasonal patterns
- Demonstrates practical AI for public health — a high-impact real-world application

JavaScript ECC Cryptography Library

Stack: Vanilla JavaScript, BigInt (zero external dependencies)

A from-scratch implementation of Elliptic Curve Cryptography on the secp256k1 curve — the same curve used in Bitcoin. Core modules include ecc.js (core EC math) and hmac.js. This is an unusually advanced exercise for an undergraduate student, requiring deep understanding of modular arithmetic, point operations, and cryptographic protocol design.

- secp256k1 elliptic curve operations fully implemented
- HMAC and SHA-256 implementations
- Part of a larger skillsShare Node.js app with hand-rolled RSA-2048 and ECC
- No external cryptography libraries used — pure BigInt arithmetic

InstructPrune — Adaptive Visual Token Pruning for LLaVA

Stack: Python, PyTorch, LLaVA-1.5-7B, spaCy

A training-free method that adapts visual token pruning ratios to instruction complexity in large vision-language models. Targets CVPR 2027 Workshop on Efficient Large Vision Models.

- 3 complexity scorers: token_count, dep_depth (spaCy), attn_entropy (LLM attention entropy)

- Linear mapper from complexity score to pruning ratio
- Core model code complete; evaluation pipeline in progress
- Two known critical bugs under active debugging: incorrect L2 norm for token importance and full-sequence pruning instead of visual-token-only pruning

CrisisNet — Disaster Response Intelligence (EWU Robotics Club Hackathon)

Stack: React/Next.js, Node.js/Express, PyTorch/TensorFlow

A disaster response platform with LSTM-based flood prediction. Built with the EWU Robotics Club hackathon team.

- LSTM model for flood prediction based on temporal weather/hydro data
- Full-stack with React/Next.js frontend and Express backend
- Real-time disaster response intelligence use case

6. Research Work

ExplainFake — Primary Undergraduate Thesis

Hirok's main long-term research project is ExplainFake, a multimodal deepfake detection system that fuses audio-visual signals with explainability built directly into the architecture (not post-hoc).

- Architecture: Pretrained visual + audio encoders feeding into a cross-attention layer; attention weights serve as built-in XAI providing causal attribution, counterfactual reasoning, and uncertainty decomposition
- Dataset: FakeAVCeleb with cross-dataset testing for generalization evaluation
- Selected model repositories: EDVD-LLaMA, MM-DDL, MMDF-Net, RECCE, SIDA
- Generators: SimSwap, V-Express; Detectors: DiMoDif, AVH-Align
- Benchmark dataset: AV-Deepfake1M
- Status: Paper-reading and architecture planning stage; reading AVAD, AVFF, contextual cross-modal attention, XAI evaluation surveys
- Publication target: Computers in Biology and Medicine (Q1 journal)

InstructPrune (Research Project)

- Training-free adaptive visual token pruning for LLaVA-1.5-7B
- Core idea: pruning ratio adapts to instruction complexity — simple question = aggressive pruning, complex = conservative
- 3 complexity scorers: ``token_count``, ``dep_depth`` (spaCy), ``attn_entropy`` (LLM attention entropy) → linear mapper to pruning ratio
- Target venue: CVPR 2027 Workshop on Efficient Large Vision Models
- Core code done (``src/model/``, ``src/instructprune/``, ``src/baselines/``)
- Eval pipeline NOT built yet (``benchmark_loader``, ``evaluator``, ``metrics``)
- Two critical bugs: (1) ``patched_llava.py`` uses L2 norm instead of attention weights for token importance — invalidates results; (2) hook prunes full sequence instead of isolating visual tokens only

IEEE-Style Computer Vision Survey Paper

A writing exercise producing an IEEE-formatted survey paper on a computer vision topic — demonstrating familiarity with academic writing conventions and literature organization.

ORCID Profile

Hirok maintains an ORCID profile (0009-0009-3782-5632) and a ResearchGate profile, indicating a deliberate effort to establish an academic identity early in his career.

7. Hackathons & Competitions

NRF26 National Hackathon	DengueScope — climate-driven dengue outbreak prediction platform for Bangladesh
Global Fusion Hackathon 2026	CropGuard AI — EfficientNet crop disease detection + Groq/Llama 3 guidance + PostGIS heatmap
EWU Robotics Club Hackathon	CrisisNet — LSTM-based flood prediction + disaster response intelligence
Infinity AI BuildFest	Active participation (cloudcampbd.com/the-infinity-ai-buildfest)
Luma Hackathon	Active participation (luma.com event)
RoboSub (BRACU DUBURI)	International AUV competition via BRACU DUBURI team (machine vision contributions)

8. Career Goals & Trajectory

Near-Term (2026–2027)

- Intermediate ML Engineer role by January 2027
- Complete BSc at BRAC University (expected December 2026)
- Complete MLOps Specialization certification
- Complete NLP Specialization certification
- Obtain AWS ML Associate certification

Medium-Term (2027–2029)

- Publish research in high-impact venues (Computers in Biology and Medicine Q1, CVPR Workshop)
- Build strong research publication record for graduate applications
- Establish AI automation agency (Korviora) as a side venture

- Pursue graduate studies with scholarship — aspirational targets include Stanford and top research universities

Long-Term

- AI Researcher at a top-tier research lab (e.g., Google DeepMind)
- Postgraduate research in data science / AI at a world-leading institution
- Potential launch of EHR (Electronic Health Record) system and health AI products

9. Online Presence & Public Profile

Hirok maintains an active, multi-platform professional presence. His GitHub account (github.com/hirokr) has 68 repositories and has earned four GitHub achievements: Pair Extraordinaire, Pull Shark (x2), YOLO, and Quickdraw — reflecting active collaborative development. His bio reads: "Making something from the scratch is fun."

Portfolio	hirokrr.vercel.app
GitHub	github.com/hirokr — 68 repos, 6 followers, 34 following
LinkedIn	linkedin.com/in/hirokrr — self-described as Aspiring AI Researcher
ResearchGate	researchgate.net/profile/Hirok-Roy-Rahul
ORCID	orcid.org/0009-0009-3782-5632
Kaggle	kaggle.com/hirokrr
LeetCode	leetcode.com/hrrahul
Codeforces	codeforces.com/profile/hrrahul0625
HackerRank	hackerrank.com/hirokrr
AUV Team Email	hrahul@bracu-duburi.com

10. Academic Coursework & Assignments

11. Additional Interests & Activities

AI Automation Agency — Korviora

Hirok is planning an AI automation agency under the brand name Korviora, focused on AI chatbots, e-commerce automation, and customer support automation. This reflects his interest in productizing AI beyond research into commercial applications.

12. Analytical Summary

Strengths

- Exceptional breadth for a final-year undergraduate — spans production web development, embedded systems (AUV), cryptography, ML research, and AI engineering simultaneously
- Systems-first thinker who consistently chooses architecture-heavy solutions over quick hacks
- Strong research ambition backed by concrete output: real projects targeting Q1 journals and CVPR workshops
- Practical zero-budget engineering mindset — maximizes capability from free-tier infrastructure
- Demonstrated ability to deliver under hackathon pressure across multiple competitions
- Rare combination of ML research depth + production full-stack skills + hardware-adjacent AUV work
- Active public portfolio with ORCID, ResearchGate, and GitHub — professional identity well-established early

Risks & Challenges

- Fragmentation risk: simultaneously pursuing advanced AI research, multiple startups, web/mobile engineering, automation, competitive programming, and networking — execution depth may suffer
- Several flagship projects (ExplainFake, InstructPrune) are at early stages with known technical bugs unresolved — completion and publication timelines are uncertain
- Gap between ambition and execution validation: many ideas are conceived at startup scale before smaller MVPs are validated
- English as a second language may present friction in academic writing and international networking
- Heavy reliance on free-tier/zero-budget infrastructure may limit scalability of deployed projects

Overall Assessment

Hirok Roy Rahul is an unusually high-initiative final-year CSE student who has positioned himself at the intersection of AI research, product engineering, and academic computer science. His profile is significantly more developed than a typical undergraduate's — featuring real professional experience, international competition participation, research projects targeting top venues, and a substantial public technical footprint. The primary challenge going forward is prioritization: the ambition that drives his breadth is also the main threat to execution depth. If he can focus his energy on one or two flagship outputs in 2026 — completing ExplainFake or InstructPrune to publication quality, or shipping one production-grade product — he will have a compelling case for top graduate programs and senior ML roles alike.